Alejandro Gomez Espinosa, Ph.D.

Postdoctoral Research Associate & FNAL LPC Junior Distinguished Researcher
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Research Profile

I am an experimental high-energy physicist specializing in Higgs physics and advanced reconstruction algorithms within the CMS Collaboration. My research focuses on exploring the Higgs self-coupling through di-Higgs production in the four-b-quark ($b\bar{b}b\bar{b}$) and $b\bar{b}W^+W^-$ ($bbWW$) decay channels, leveraging a strong background in searches for exotic new physics. As a leader in jet reconstruction and workflow preservation groups, I have directed the development of critical calibration tools and analysis reproducibility workflows to accelerate scientific discovery.

Education

Rutgers, The State University of New Jersey — **May 2018**
Ph.D., Experimental High Energy Physics — *New Brunswick, USA*
Thesis topic: Search for pair-produced diquark resonances in proton-proton collisions with the CMS detector at $\sqrt{s} = 13$ TeV
Advisor: Prof. Eva Halkiadakis

Escuela Politecnica Nacional — **August 2012**
B.Sc. Physics — *Quito, Ecuador*
Research Project: Search for Heavy Scalar Bosons decaying into Four Tops at the LHC.
Advisors: Prof. Edgar Carrera (USF-Q), Prof. Francisco Yumiceva (Florida Tech).

Significant publications and public results

A full list of publications can be found in the following INSPIRE link.

Primary author

- CMS Collaboration. *Improved results on Higgs boson pair production in the 4b final state* CMS-PAS-HIG-24-011. (Public twiki) *Submitted to Physical Review D*
  Contribution: Executed the complete Run 2 analysis chain and performed the statistical combination with Run 3 studies.
- CMS Collaboration. *Search for high transverse momentum $H \to b\bar{b}$ produced in association with semileptonic $t\bar{t}$ at $\sqrt{s} = 13$ TeV.* (Co-supervisor) *Ph.D. thesis* CMS internal link (AN/2020-015).
  Contribution: Co-supervised the Ph.D. student and designed the overall analysis strategy.
- CMS Collaboration. *Search for pair-produced resonances decaying to quark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV*. Phys.Rev.D **98**, 112014. 2018. (Public twiki)
  Contribution: Sole primary analyst responsible for the design, execution, and paper preparation, integrating four distinct search topologies.

- CMS Collaboration. *Search for low-mass pair-produced dijet resonances using jet substructure techniques in proton-proton collisions at* $\sqrt{s} = 13$ *TeV*. CMS Physics Analysis Summary CMS-PAS-EXO-16-029. (Public twiki)
  Contribution: Primary analyst responsible for the entire search lifecycle, including the design and implementation of the first CMS jet substructure triggers.

Contributing author

- CMS Collaboration. *Performance of heavy-flavour jet identification in boosted topologies in proton-proton collisions at 13 TeV* CMS-PAS-BTV-22-001
  Contribution: Derived calibrations and scale factors for boosted heavy-flavor jets using the muon-tagging method.
- CMS Collaboration. *Measurement of the ttH and tH production rates in the Hbb decay channel using proton-proton collision data at* $\sqrt{s} = 13$ *TeV.* JHEP 02 (2025) 097
  Contribution: Developed discriminant variables to separate signal from background processes, and performed high transverse momentum studies.
- CMS Collaboration. *Inclusive search for the standard model Higgs boson produced in pp collisions at* $\sqrt{s} = 13$ *TeV using* $H \to b\bar{b}$ *decays*. Phys. Rev. Lett. **120**, 071802. 2018. (Public twiki).
  Contribution: Optimized and measured the efficiency of jet substructure triggers.
- CMS Collaboration. *Search for light vector resonances decaying to a quark pair produced in association with a jet in proton-proton collisions at* $\sqrt{s} = 13$ *TeV*. JHEP 01 097. 2018. (Public twiki).
  Contribution: Optimized and measured the efficiency of hadronic triggers.
- CMS Collaboration. *Search for low mass vector resonances decaying to quark-antiquark pairs in proton-proton collisions at* $\sqrt{s} = 13$ *TeV*. Phys. Rev. Lett. **119**, 111802. 2017. (Public twiki).
  Contribution: Optimized and measured the efficiency of boosted jet triggers.

Professional service

Employment positions

Carnegie Mellon University | **Oct 2023 – Present**
Postdoctoral Research Associate | *Department of Physics*

- **Co-leading** the CMS di-Higgs to four b-quark ($b\bar{b}b\bar{b}$) analysis group, while developing a novel analysis strategy for the $bbWW$ channel utilizing low-p_{T} jets.
- Developed the Run 2 $b\bar{b}b\bar{b}$ search using data-driven background estimations and synthetic datasets, **improving sensitivity by 30%**; fully preserved the workflow using Snakemake and REANA (120+ jobs).
- Designed and deployed multiple continuous integration (CI) pipelines (up to 50+ jobs) to automate software validation and ensure code stability across collaborative repositories.
- Organized and instructed multiple training sessions on workflow orchestration and analysis preservation for the wider CMS community.

ETH Zurich | **Oct 2022 – Sep 2023**
Visitor Researcher | *IPA*

- Developed and implemented continuous integration (CI) pipelines to automate the validation of jet substructure software releases.
- Finalized data measurements and documentation for jet substructure observables in high transverse momentum (p_{T}) environments.

ETH Zurich **April 2018 – Sept 2022**
Postdoctoral Researcher *IPA*

- Key contributor to the Higgs boson measurement in the $t\bar{t}H(b\bar{b})$ channel; co-led the boosted-regime analysis team within the local group and collaborated closely with 5 international research institutions.
- Led jet substructure measurements in different high transverse momentum (p_{T}) jet topologies, managing the analysis alongside graduate students and publishing public physics results.
- Supervised and mentored graduate and undergraduate students working on jet substructure observables and the commissioning of boosted heavy-flavour identification taggers.
- Contributed to the commissioning of boosted taggers for heavy resonances and pileup mitigation techniques at the High-Level Trigger for Run 3.

Rutgers, The State University of New Jersey **June 2013 – March 2018**
Graduate Research Assistant *Department of Physics*

- Primary analyst for the search for pair-produced dijet resonances using jet substructure techniques; designed and implemented the first CMS jet substructure triggers.
- Supervised and mentored undergraduate students, Research Experience for Undergraduates (REU) participants, and high school students within the research group.

Leadership positions

CMS Experiment, Common Analysis Tools Group **Sept 2024 – Present**
Workflow Orchestration and Analysis Preservation Group Convener

- Directing the CMS Experiment efforts to promote workflow orchestration tools, continuous integration practices, and analysis preservation.
- **Created standard Snakemake and REANA tutorials** for High Energy Physics, establishing educational resources now adopted and taught across multiple training schools.

CMS Experiment, JetMET Group **Sept 2019 – Sept 2021**
JetMET Algorithms and Reconstructions (JMAR) Group Convener

- Chaired the coordination of 10+ sub-groups and teams validating jet reconstruction, substructure, and pileup mitigation algorithms.
- Directed overall CMS implementation, validation, and release of jet and MET calibration tools for Run 2.

CMS Experiment, Higgs Group **July 2018 – April 2021**
JetMET-HIG Contact

- Served as the liaison between the JetMET object groups and the Higgs physics group, reviewing jet and MET reconstruction applications in all Higgs analyses.
- Reported alignment issues and advised analyzers dealing with complex jet/MET uncertainties.

CMS Experiment, Beyond Two Generations (B2G) Group **Sept 2016 – Sept 2017**
B2G-Trigger Contact

- Acted as the liaison between trigger coordination and the B2G physics group, studying the performance and efficiency of triggers across all B2G analyses.
- Assisted analysts in trigger development and reviewed trigger performance studies for group approvals.

CMS Experiment, BRIL Group **April 2015 – April 2017**
BRIL-DQM Contact

- Liaison between Data Quality Monitoring (DQM) and the Beam, Radiation, Instrumentation and Luminosity (BRIL) groups.
- Developed a new framework within the central CMS DQM system to monitor luminometer performance online.

Awards and Honors

LHC Physics Center at Fermilab **January 2026**

2026 Junior CMS LPC Distinguished Researcher

Fellowship provides recognition and support to early career researchers who conduct their own independent research at the LHC Physics Center (LPC) at Fermilab. Award includes partial salary support for one year and travel support to attend CMS meetings and conferences.

Breakthrough Prize Foundation **April 2025**

Breakthrough Prize in Fundamental Physics

Awarded for significant contributions to the understanding of the Higgs boson and its properties, along with my colleagues in the CMS Collaboration.

American Physics Society (APS) - Division of Particles and Fields (DPF) **April 2016**

Student Travel Award

Awarded travel funds to present my contributions to the search for paired dijet resonances in the boosted regime at the APS DPF Meeting 2017.

CERN Summer Students Programme **June 2012 – August 2012**

CERN Summer Student Fellowship

Selected for the prestigious CERN Summer Student Fellowship. Researched selection criteria improvements for the Higgs boson search decaying to W^+W^- in the leptonic channel under the supervision of Prof. Lawrence Sulak (Boston University).

LHC Physics Center at Fermilab **November 2011 – May 2012**

Visitor Undergraduate Scientist Fellowship

Awarded a visitor fellowship to conduct research at the LHC Physics Center (LPC) at Fermilab. Served as the primary contributor to the search for heavy scalar bosons decaying into four top quarks as part of my B.Sc. thesis under the supervision of Prof. Francisco Yumiceva (Florida Tech).

Analysis Review Committee

- CMS Experiment, ”CMS Analysis Frameworks”, CAT-24-001. *Ongoing*.
- CMS Experiment, ”Particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons”, Submitted to JHEP.
- CMS Experiment, ”Search for Higgs boson decay to a charm quark-antiquark pair in proton-proton collisions at 13 TeV”, Phys. Rev. Lett. 131, 061801.
- CMS Experiment, ”Inclusive search for a boosted Higgs boson decaying to charm quark pairs in proton-proton collisions at 13 TeV”, Phys. Rev. Lett. 131, 041801.

Workshop organizer

- CAT Hackathon. LPC Fermilab. February 2026. Organizer.
- JetMET Hackathon. LPC Fermilab. January 2026. Organizer.
- EPIC-III Scientific Computing School. August 2023. (In Spanish) Quito-Ecuador. Main organizer and contributor for the particle physics exercise.
- EPIC-I Scientific Computing School. October 2021. (In Spanish) Quito-Ecuador. Main organizer and contributor for the particle physics exercise.

Committees

- USCMS Mentorship Program, Committee Member. 2024–present.

Student Supervision & Mentorship

- *Aniket Khanal* (CMU) Co-supervised *Ph.D. research*. Search for non-resonant di-Higgs production decaying into $bbWW$ using low-p_{T} jets. (2024–Present)
- *Sindhu Murphy* (CMU) Co-supervised *Ph.D. research*. Search for resonant di-Higgs production decaying into four b-quarks. (2024–Present)
- *Pamela Llerena* (EPN-Quito) Co-supervised *B.Sc. thesis project*. Study of the four-top signature in the μ+jets channel using CMS Open Data. (2023)
- *Andres Chicaiza* (EPN-Quito) Co-supervised *B.Sc. thesis project*. Study of the four-top signature in the e+jets channel using CMS Open Data. (2023)
- *Matteo Marchegiani* (ETH Zurich) Co-supervised *Ph.D. thesis*. Search for high-p_{T} $H \to b\bar{b}$ produced in association with dileptonic $t\bar{t}$ at $\sqrt{s} = 13$ TeV. (2020–2022)
- *Daniele Ruini* (ETH Zurich) Co-supervised *Ph.D. thesis*. Search for high-p_{T} $H \to b\bar{b}$ produced in association with semileptonic $t\bar{t}$ at $\sqrt{s} = 13$ TeV. (2019–2021)
- *Kaustuv Datta* (ETH Zurich) Mentored *Master's thesis*. Novel jet observables for the identification and calibration of boosted hadronic W decays. (2019)
- *Geliang Liu* (ETH Zurich) Mentored *Master's project*. Unfolding of jet substructure variables in a dijet-enriched environment using 2016 CMS data. (2020)
- *Diego Coloma* (USFQ-Quito) Mentored *undergraduate project*. Implementation of pileup mitigation techniques in the High-Level Trigger of the CMS Experiment. (2020)
- *Jean Somalwar* (Rutgers University) Mentored *undergraduate project*. Background estimation studies for pair-dijet searches with boosted jets. (2015–2017)
- *Alan Kahn* (Rutgers University) Mentored *undergraduate project*. Efficiency studies for substructure triggers. (2014–2016)
- *Margaret Morris* (Rutgers University) Mentored *undergraduate project*. Optimization studies for pair-dijet searches with resolved jets. (2014–2016)

Teaching Experience

CMS Data Analysis School (CMSDAS) **Jan 2026, Jan 2025, Jan/July 2024, Jan 2021, Sept 2020**
Facilitator
Main contributor and instructor for the jet exercise at the CMS Data Analysis School (CMSDAS) for graduate-level newcomers. Developed and taught a new introductory tutorial on workflow reinterpretation and analysis preservation using REANA.

ETH Zurich **September 2018 – September 2022**
Teaching Assistant
General physics laboratory instructor for science undergraduates. Teaching assistant for the Statistical Methods in Particle Physics course in the Master of Science in Physics program.

Rutgers, The State University of New Jersey **September 2012 – May 2013**
Teaching Assistant
Laboratory instructor for introductory college physics courses.

San Gabriel High School **January 2011 – March 2012**
High School Teacher *Quito, Ecuador*
Taught physics and mathematics to senior high school students.

Conferences and talks

Plenary & Invited Talks

- HIGGS 2025. *Experimental methods in Higgs measurements at ATLAS and CMS*. On behalf of ATLAS and CMS. October 2025.
- Semivisible Jets Workshop 2022. *Jet in CMS*. July 2022.
- BOOST 2021. *Jet substructure measurements in CMS*. On behalf of CMS. August 2021.
- SM@LHC 2021. *Jet substructure at the LHC*. On behalf of ATLAS, CMS and LHCb. April 2021.
- XIIth International Workshop on the Interconnection between Particle Physics and Cosmology. *Exotic searches at the LHC*. Zurich, Switzerland. August 2018.

Contributed Talks

- USCMS Meeting 2025. *Preproducible analysis in CMS using REANA*. Houston, USA. May 2025.
- 9th Workshop: (Re)interpretation of the LHC results for new physics. *CMS Analysis Preservation efforts*. Geneva, Switzerland. February 2025.
- SILAFAE 2022. *Measurements of Higgs boson decaying into heavy flavored quarks in CMS*. Quito, Ecuador. November 2022.
- ML4Jets 2020. *Jet substructure tagging and pileup mitigation at CMS*. New York, USA. January 2020.
- 31st Reencontres de Blois. *Searches for boosted objects at ATLAS and CMS*. Blois, France. June 2019.
- SUSY 2018. *Searches for R-parity violating supersymmetry with CMS*. Barcelona, Spain. July 2018.
- APS DPF Meeting 2017. *Search for paired dijet resonances in the boosted regime with the CMS detector at 13 TeV*. Fermilab, Batavia, USA. August 2017.
- APS April Meeting 2016. *CMS Run-2 Instrumentation for beam radiation and luminosity measurement using novel detector technologies*. Utah, USA. April 2016.

Seminars

- What are we doing at the LHC and what is the Ecuadorian contribution. School of Physics and Nanotechnology. Yachay Tech. Urcuqui, Ecuador. Jun 2023.
- El Gran Collisionador de Hadrones, la Inteligencia Artificial y la experiencia Ecuatoriana. Department of Physics Seminar. EPN. Quito, Ecuador. February 2023.

Continuous training

Deep Learning Specialization **2021**
Deep Learning.AI - Coursera
Online specialization in deep learning techniques: deep neural networks, hyperparameter optimization, convolutional neural networks, sequence models.

Google Data Analytics Professional Certificate **2022**
Google - Coursera
Online specialization in data analytics, data cleaning and visualizations.

Open Science & Software Development

- coffea: A columnar object framework for CMS analysis. Contributions to jet and MET tools, enabling high-performance analysis on large-scale datasets.
- REANA: Reusable analyses framework for reproducible research. Developed slurm backends and Snakemake workflow integration for distributed HEP environments.
- **REANA Cluster Deployment**: Configured and deployed a local REANA cluster on Carnegie Mellon University (CMU) computing infrastructure to support analysis preservation. Integrated custom features, resolved deployment bottlenecks, and optimized resource scheduling for physics workflows.

Tutorials & Training Material

- REANA for CMS: Tutorial on analysis preservation and reproducibility using REANA in the CMS experiment.
- Snakemake for HEP: Tutorial on workflow orchestration using Snakemake for High Energy Physics analyses.

Outreach

Latitud Cero Group **January 2020-present**
Founding Member
Outreach projects and talks for Ecuadorian physics undergraduates. (In Spanish)

CERN Spanish High School Teachers Program **July 2019/2020**
Lecturer
Introduction to Experimental Particle Physics. (In Spanish)

International Masterclasses **March 2023**
Organizer
CERN-Fermilab video conferences for Ecuadorian High School Students.

International Masterclasses **March 2016/2017/2019/2020/2021/2022**
Moderator
CERN video conferences with high school students from all over the world to answer questions, encouraging them to pursue a scientific career.

CMS Voices **June 2016**

Account Manager

Managed the CMS Voices Twitter account for a month to share the daily life of a particle physicist, answer public questions, and promote engagement with physics research.

Extra activities

Enologist	Specialization in Enology (Winemaking). Euroinnova Business School. 2022.
Peregrino	Completed an 800 km pilgrimage walk along the Camino de Santiago, Spain. September 2022.
Sport Performance Analytics	Specialization in Sports Performance Analytics. Coursera. 2021.

Languages

Spanish	Native
English	Fluent
German	Basic
Italian	Basic
French	Basic